



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 64: TI Solar Battery Pack Bi-Weekly Update 5

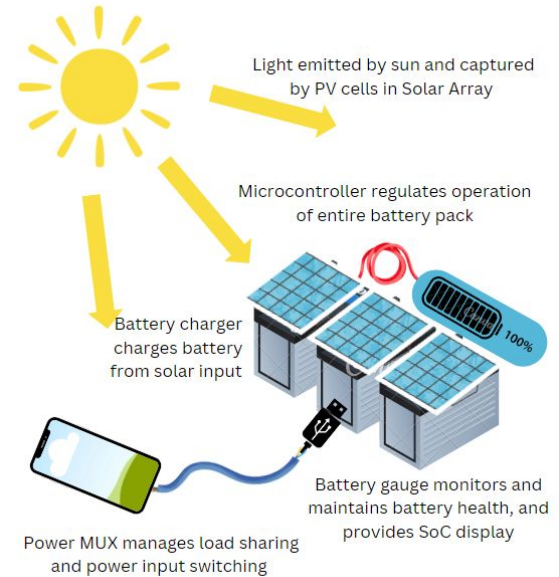
Jaein Cha, Patrick Wang, Cole Miller, Camden Pomeroy

Sponsor: Wyatt Keller

TA: Ali Alenezi

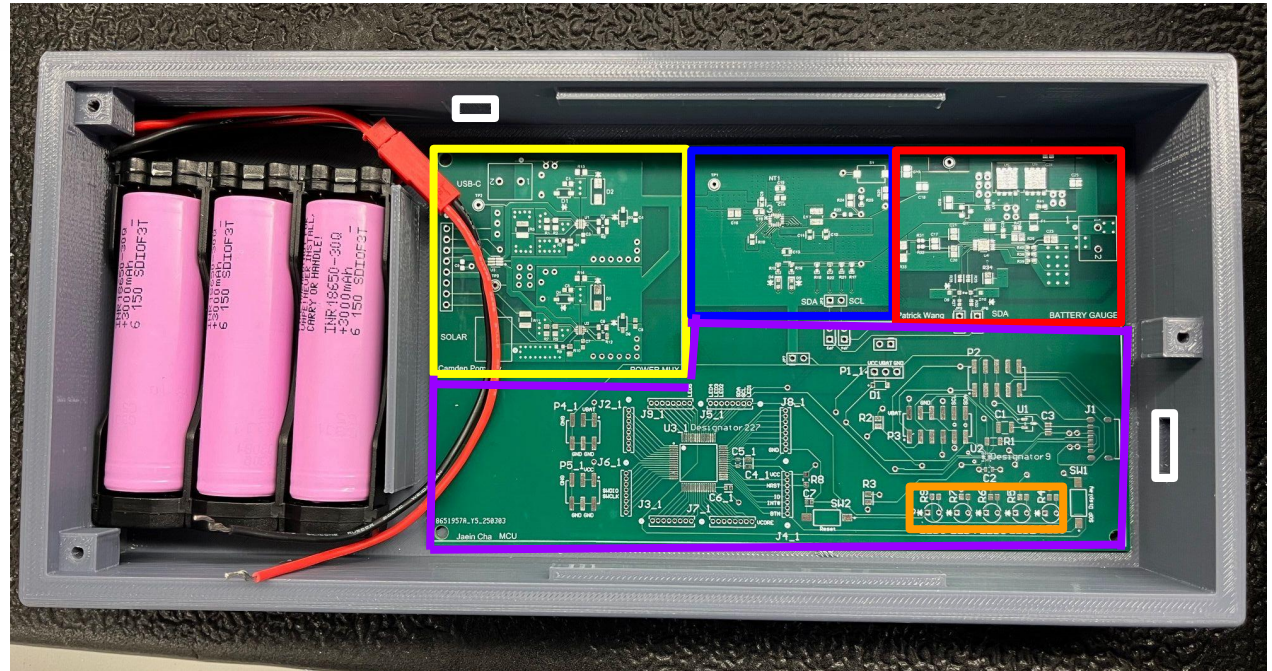
Project Summary

- Natural disasters can cause unexpected power outages resulting in the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.

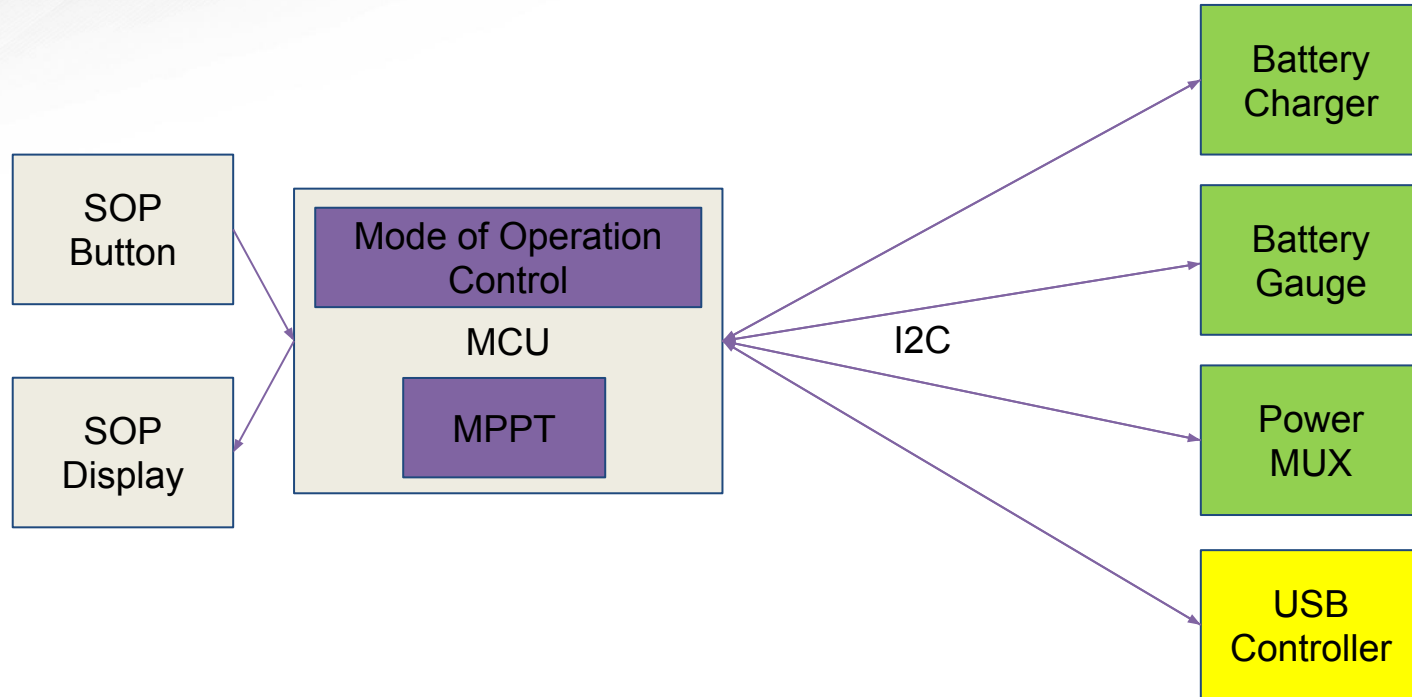


Integrated System Diagram

Yellow - Power Mux
Blue - Battery Charger
Red - Battery Gauge
Purple - MCU
Orange - SOC Display
White - Power Ports



Integrated System Diagram (software)





Project Timeline

Subsystem Designs and Testing (completed 2/3)	Integration of High Power Path (completed 2/17)	Final Integration (completed 3/12)	System Test (to complete by 3/31)	Validation (to complete by 4/7)	Demo and Report (to complete by 4/17)
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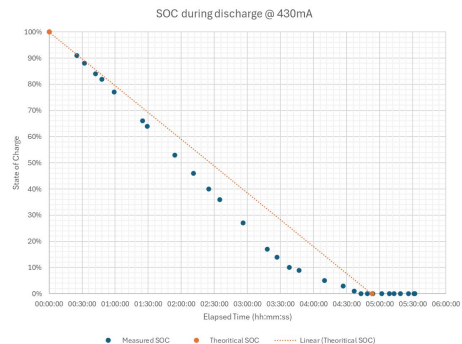
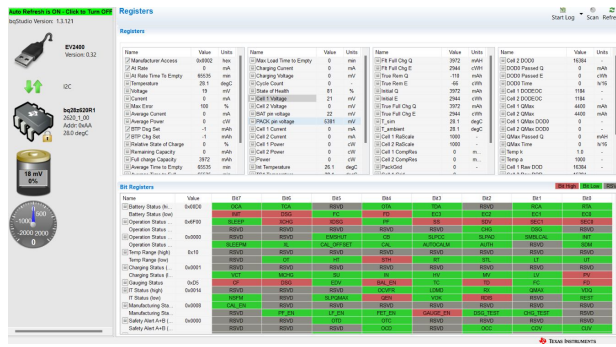
Patrick Wang

Accomplishments since last update
15 hrs of effort

- Fixed Battery Gauge I2C communication issues
- Validated correct output voltages and behavior for when V_{in} is within threshold
- Programmed gauge to calibrate temps, threshold voltages, and battery parameters

Ongoing progress/problems and plans until the next presentation

- Measure time to fully charge and discharge battery pack

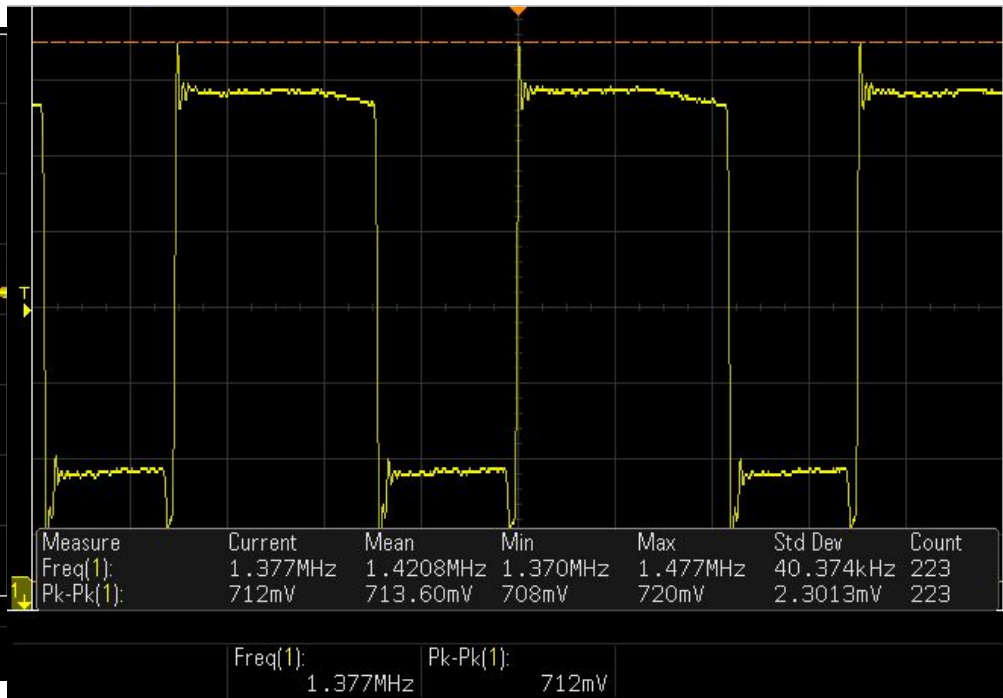
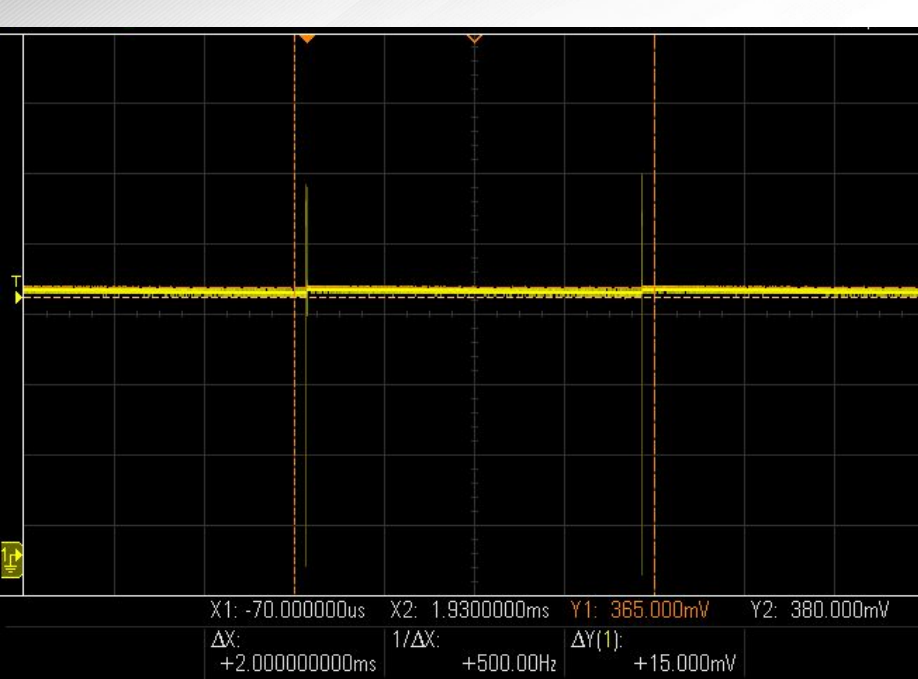




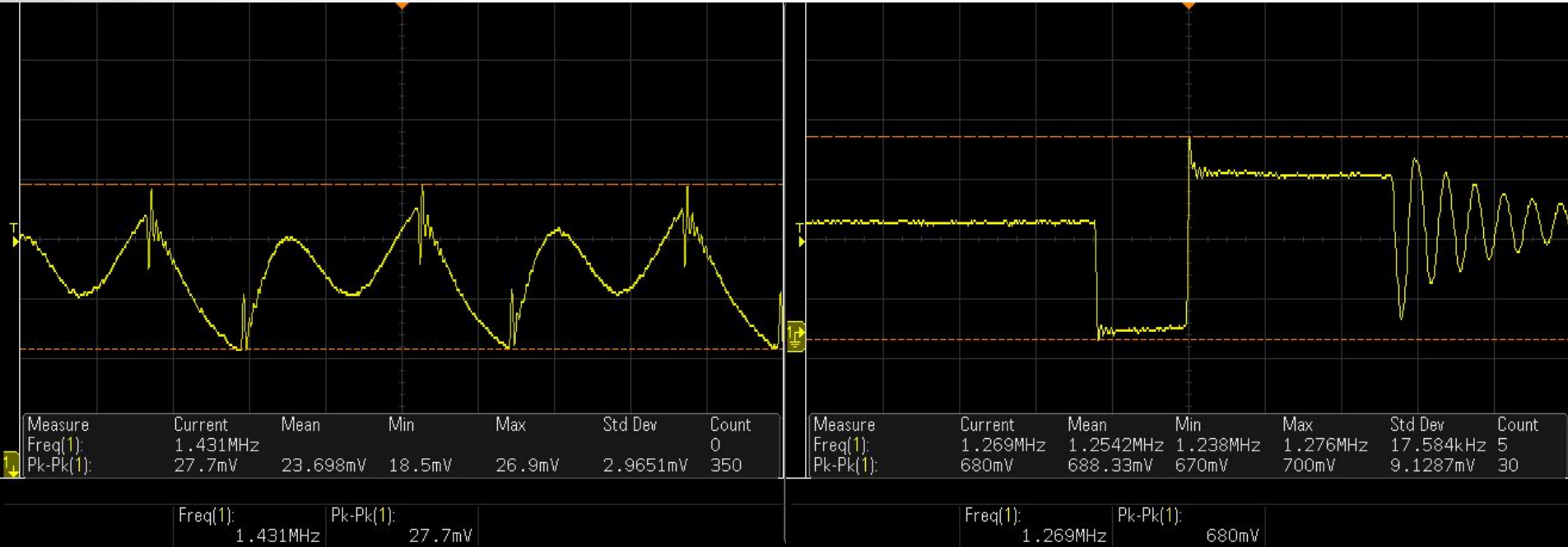
Cole Miller

Accomplishments since last update 30 hrs of effort	Ongoing progress/problems and plans until the next presentation
Fixed battery Gauge Issues Validated high power functionality Finishes assembly Validated 0.5 to 3 amp charging Validated OV and OC protections	Diagnose Boost converter waveform Characterize efficiency for full input range

Cole Miller - Buck Verification



Cole Miller - Boost Verification



Charger Operation

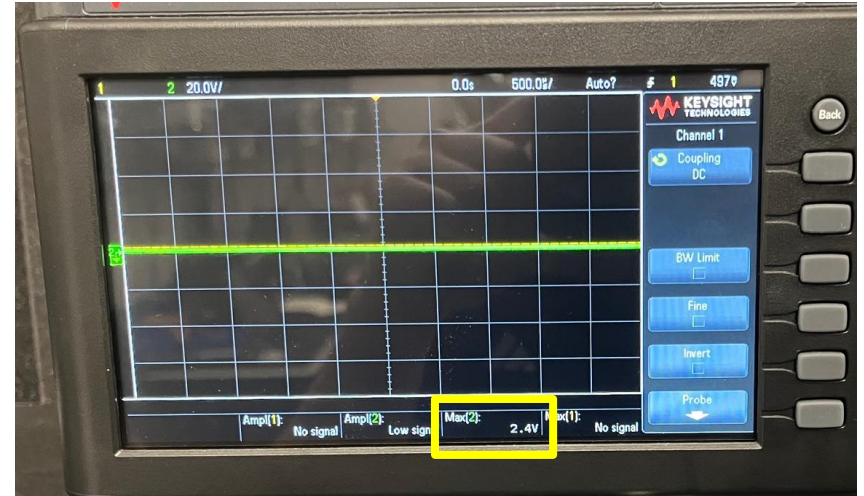
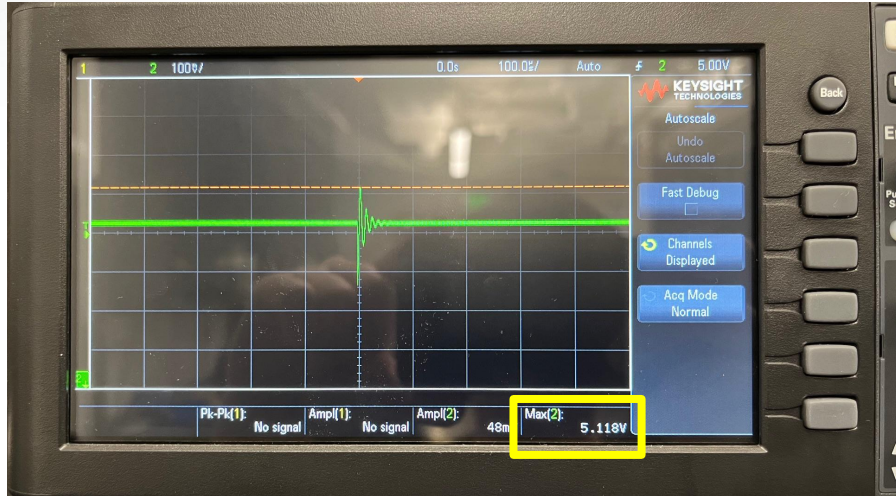




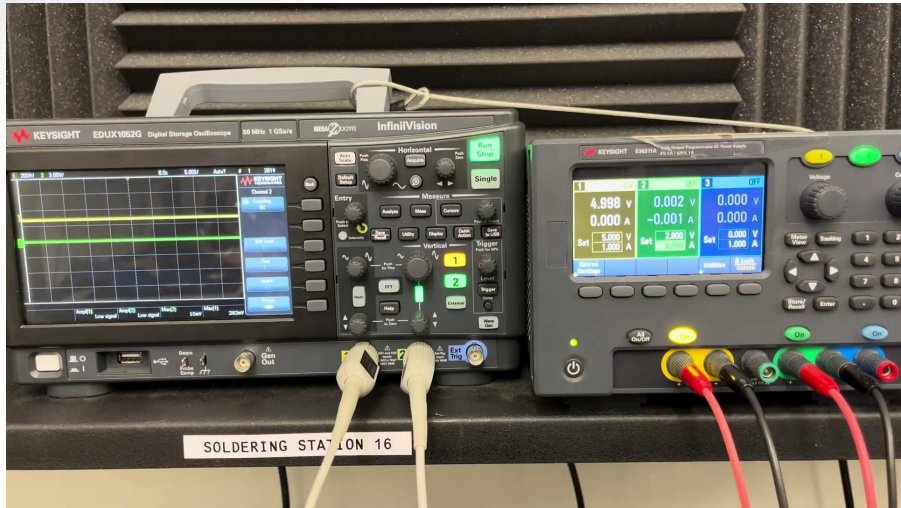
Camden Pomeroy

Accomplishments since last update 35 hrs of effort	Ongoing progress/problems and plans until the next presentation
MCU System Assembly on PCB Validated Safety of Multiple Loads Validated Charge/Discharge under wide range of solar panel operating voltage conditions Validated system shutdown during over/undervoltage	MCU I2C Communication Assembling Enclosure Collect Measurements for Report

Camden Pomeroy



Camden Pomeroy



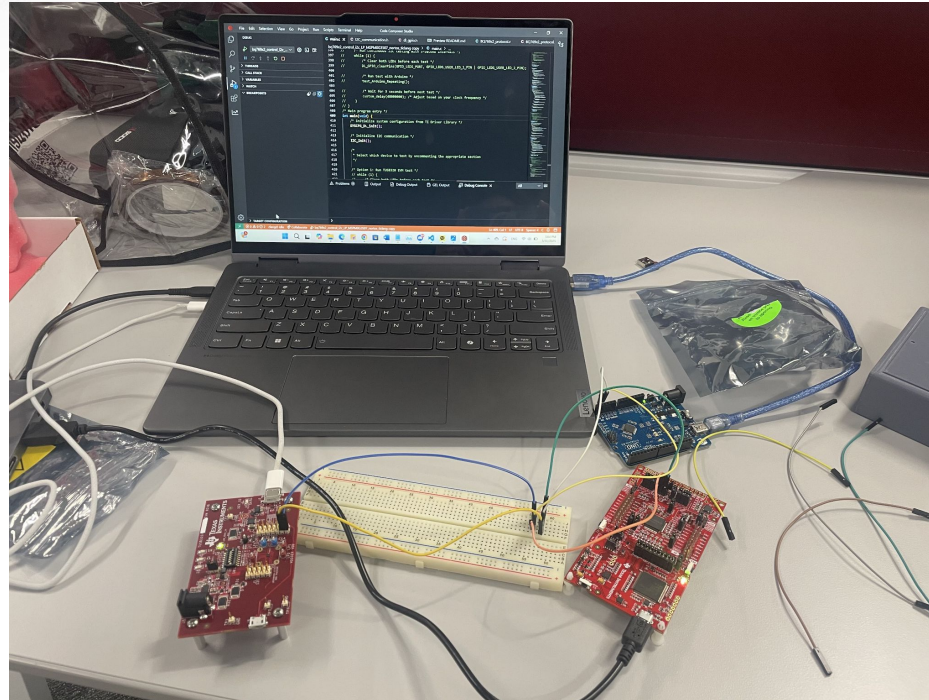


Jaein Cha

Accomplishments since last update 23 hrs of effort	Ongoing progress/problems and plans until the next presentation
- I2C serial port echoing was added for debugging purpose (additional to the LED blinking indication) - I2C debugging when multiple devices are on the bus - On board MCU connection unstable - identifying the problem for debugging	Debug the MCU connection problem On-board testing of the control logic and I2C communication Final validation



Jaein Cha



Execution Plan

[illegible]



SYSTEM Validation Plan

Number	Validation Test	Input	Expected Output	Results	Status	Engineer	Reference Number
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	As expected	Pass	Camden Pomeroy	3.2.3.2.2
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected	Pass	Camden Pomeroy	3.2.3.2.2
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested	Pass	Camden Pomeroy	6.1.1
4	PP1 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	Not tested	Pass	Camden Pomeroy	6.1.1
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is read Low from MCU	Not tested	Pass	Camden Pomeroy	6.1.1
6	PP2 PG (Bad signal)	If Vin < 2V	PG signal is read High from MCU	Not tested	Pass	Camden Pomeroy	6.1.1
7	PP1 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected	Pass	Camden Pomeroy	3.2.3.1
8	PP2 UVLO Operational	If Vin < UVLO	No output on VBUS	As expected	Pass	Camden Pomeroy	3.2.3.1
9	PP1 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected	Pass	Camden Pomeroy	3.2.3.1
10	PP2 OVLO Operational	If Vin > OVLO	No output on VBUS	As expected	Pass	Camden Pomeroy	3.2.3.1
11	SEL USBC	SEL is sent low from MCU	PP1 is active and PP2 is not	Not tested	Pass	Camden Pomeroy	6.1.4
12	SEL Solar	SEL is sent high from MCU	PP2 is active and PP1 is not	Not tested	Pass	Camden Pomeroy	6.1.4
13	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected	Pass	Camden Pomeroy	3.2.3.2.2
14	SOC Display	SOC from 0-100%	1,2,3,4, or 5 LEDs on as expected	Not tested	Pass	Jaein Cha	3.2.3.2.1
15	I2C functionality (1:1, EVM)	Run I2C test code	It reads the default register value	As expected	Pass	Jaein Cha	6.1
16	I2C functionality (1:1, on-board)	Run I2C test code	It reads the default register value	Not tested	Pass	Jaein Cha	6.1
17	I2C functionality (1:N, EVM)	Run I2C test code	It reads the default register values	As expected	Pass	Jaein Cha	6.1
18	I2C functionality (1:N, on-board)	Run I2C test code	It reads the default register values	Not tested	Pass	Jaein Cha	6.1
19	I2C - initialization	3.3V Power supply to the pc	Setup the communication correctly	As expected	Pass	Jaein Cha	6.1
20	I2C - DFP configuration	DFP plug is connected to the	Read USB I2C address and value	Not tested	Pass	Jaein Cha	6.1
21	I2C - UFP configuration	UFP plug is connected to the	Read USB I2C address and value	Not tested	Pass	Jaein Cha	6.1
22	I2C - DRP configuration	DRP plug is connected to the	Read USB I2C address and value	Not tested	Pass	Jaein Cha	6.1
23	Reset	Reset button pressed	MCU resets and stop working when	Not tested	Pass	Jaein Cha	3.2.3.2.1
24	MPPT (simulation)	Various voltages	maintain the power as close as 10	As expected	Pass	Jaein Cha	6.2
25	MPPT (on-board)	Various voltages	maintain the power as close as 10	Not tested	Pass	Jaein Cha	6.2
26	SOP Display	SOP button	Lights up all the LEDs when SOP	Not tested	Pass	Jaein Cha	3.2.3.2.1
27	Standard Operation	5V 3A on VBUS	around 5V 3A on VBUS	4.2V 3.2A on VBAT	Pass	Cole Miller	2.1
28	USB OTG Function	4.2V 3.2A on VBAT	around 4.2V 3.2A on VBAT	5V 1.5A on VBUS	Pass	Cole Miller	2.2
29	ILIM Register Functionality	If input current > 1A	output current > 1A	Output 1A	Pass	Cole Miller	2.3
30	VLM Functionality	If Vin > 4.2V	Vo > 4.2V	Output 4.2V	Pass	Cole Miller	2.4
31	Solar Panel VOC	Solar panel Voc < 18V	Solar panel Voc < 18V	As expected	Pass	Cole Miller	6.3.1
32	Battery Charger I2C	Write to registers. Write charge	Read from said registers	Not tested	Pass	Cole Miller	6.1.2
33	Solar Panel Power	regular usage in sunny day	output 15W	As expected	Pass	Cole Miller	6.3.12
34	Battery Gauge I2C	Write to registers. Read battery	Read from said registers	Not tested	Pass	Patrick Wang	6.1.3
35	Battery Lifespan	Battery connected to load	The lifespan is at least 3 hours	Not tested	Pass	Patrick Wang	3.2.1.2
36	Power Consumption	Normal Operating Condition	Maximum power consumption is 15	Not tested	Pass	Patrick Wang	3.2.1.2
37	Battery SOH	Discharge batteries from 100	Batteries stay in within an appropriate	Not tested	Pass	Patrick Wang	5.1
38	High Power Path Charge	5V 3A from power MUX	Charges Battery Pack	As expected	Pass	Patrick, Cole, Camden	3.2.3.2.2
39	High Power Path Discharge	Load Connected to USBC	Battery Pack Discharges 15W	Not tested	Pass	Patrick, Cole, Camden	3.2.3.2.2
40	USBC Charging	USBC plug from wall outlet	Battery Pack Charges	Not tested	Pass	Patrick Wang	6.2.1



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Final slide – Thank you